

DrawDiff Comparison Report

Part number: DA1840189 | Job: f56bd7eb-7cea-4150-ae35-fdc14487ae0b | Generated: 2026-04-30T02:33:43+00:00

verify3: tight Pass B + graph-aware matcher

Summary

0 Critical	21 Significant	3 Minor	0 Uncertain	4 Views compared
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Executive Summary

Rev A of DA1840189 (issued 10/16/25) introduces several significant geometry changes to this bracket/clamp block alongside a substantial drawing reorganization. Most critically, multiple fillet/radius additions are evident: slot opening interior corners on the rear face transition from sharp to radiused, the lower edge profile of the bracket gains fillets, and the left mounting boss blend geometry changes to a more pronounced S-curve transition — all of which affect stress concentration, machining toolpaths, and potentially fixture requirements. The R.030 pocket corner callout count is halved from 16X to 8X in Section C-C, requiring clarification as to whether corners were removed from the design or merely re-grouped under a different leader. The drawing set itself was substantially restructured: views VIEW_1, SIDE, and VIEW_8 were removed and replaced with a new FRONT view, Sections A-A, B-B, and C-C, and an ANNOTATION view, reflecting a more fully detailed second sheet; the revision block title field also lost its 'TS' designation, which should be confirmed as intentional. Minor documentation updates include replacing the brand-specific 'HELICOIL' callout with the generic 'HELICAL INSERTS' and updating all revision identifiers from NC to A.

Title Block

Field	Original	Revised
part_number		
revision	TS	
material		
tolerance		
drawn_by		
date		

Revision Block

Added in revised

No revision entries on revised drawing.

Removed from original

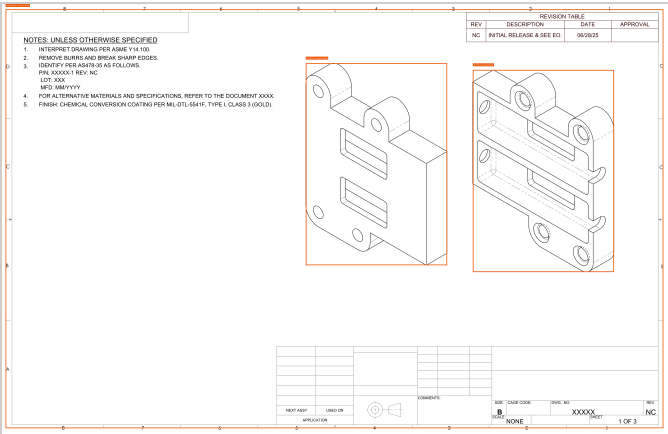
No revision entries on original drawing.

Annotated Drawings

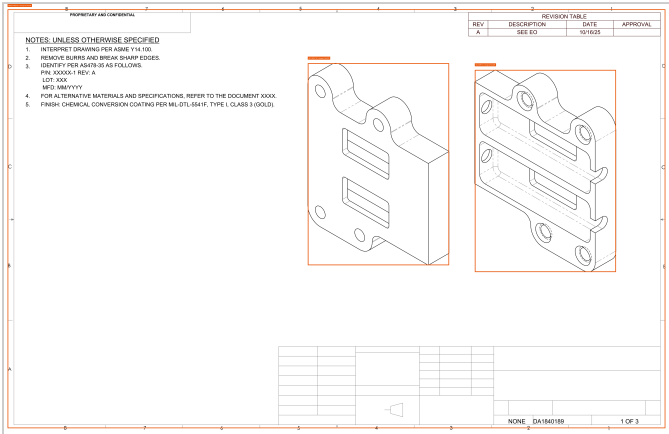
Severity: ■ Critical ■ Significant ■ Minor ■ Uncertain

Sheet 1 — boxes mark views with detected changes.

Original

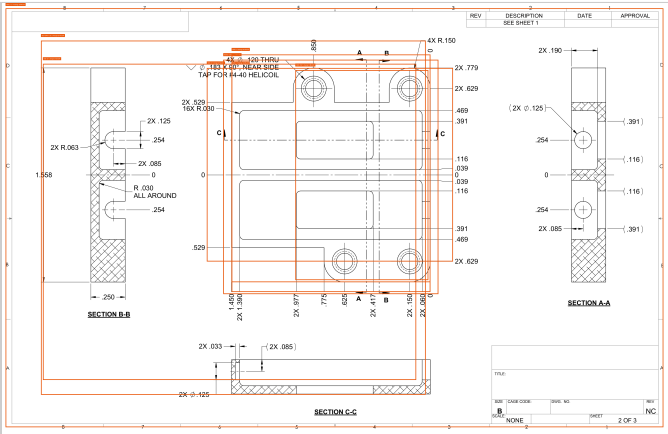


Revised

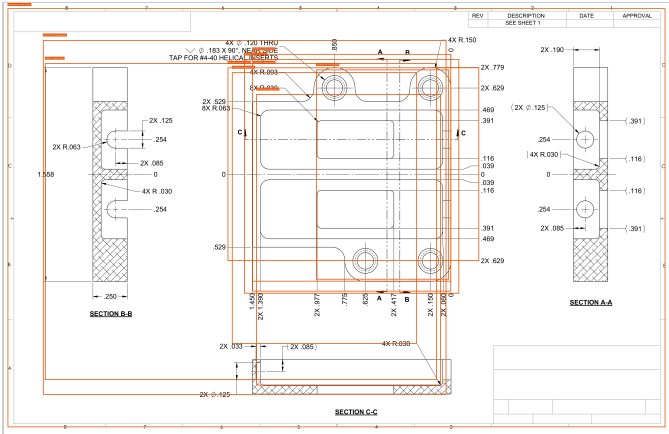


Sheet 2 — boxes mark views with detected changes.

Original



Revised



View-by-view Changes

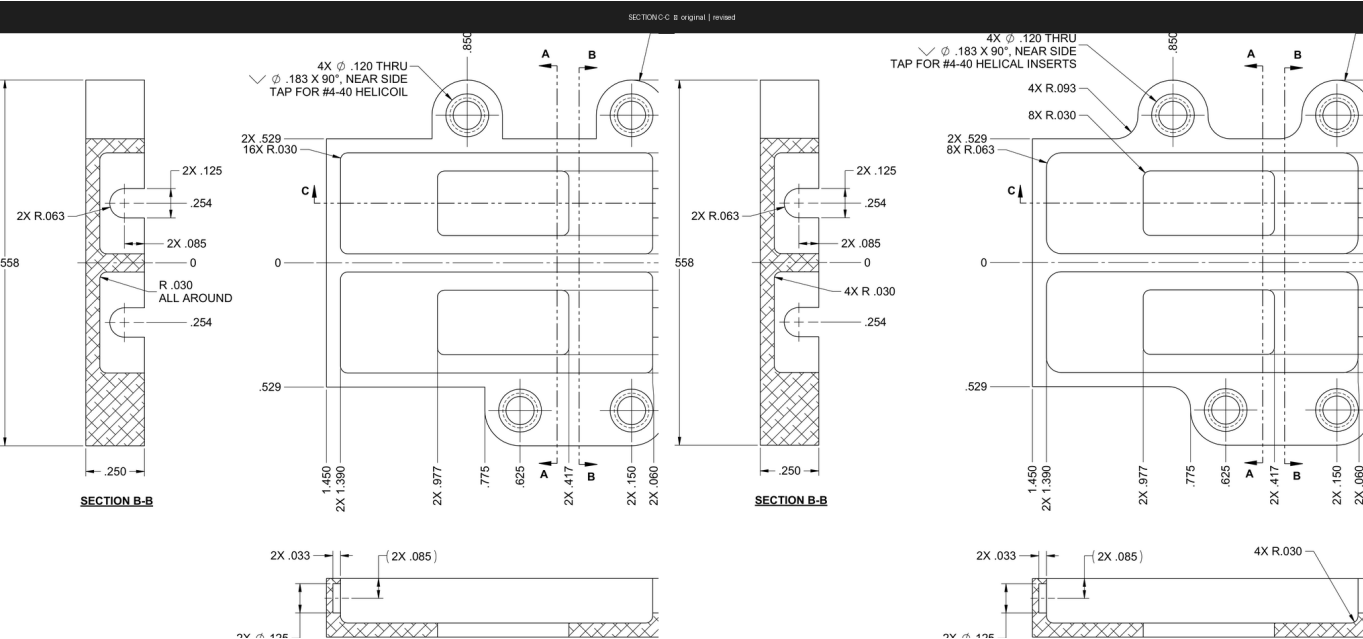
title_block [TITLE_BLOCK]

Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	revision	TS	—	1.00	title_block.revision changed from 'TS' to None Impact: Revision field in title block was cleared from 'TS' — confirm this is intentional and not a data-entry omission before release.

revision_block [REVISION_BLOCK]

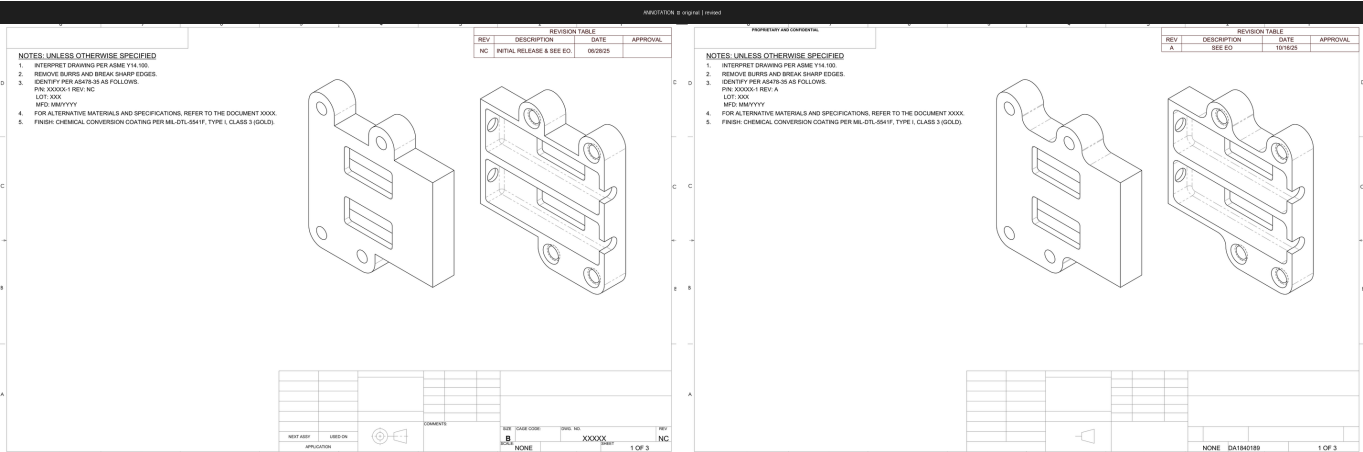
No textual changes recorded for this view.

SECTION C-C [MATCHED]

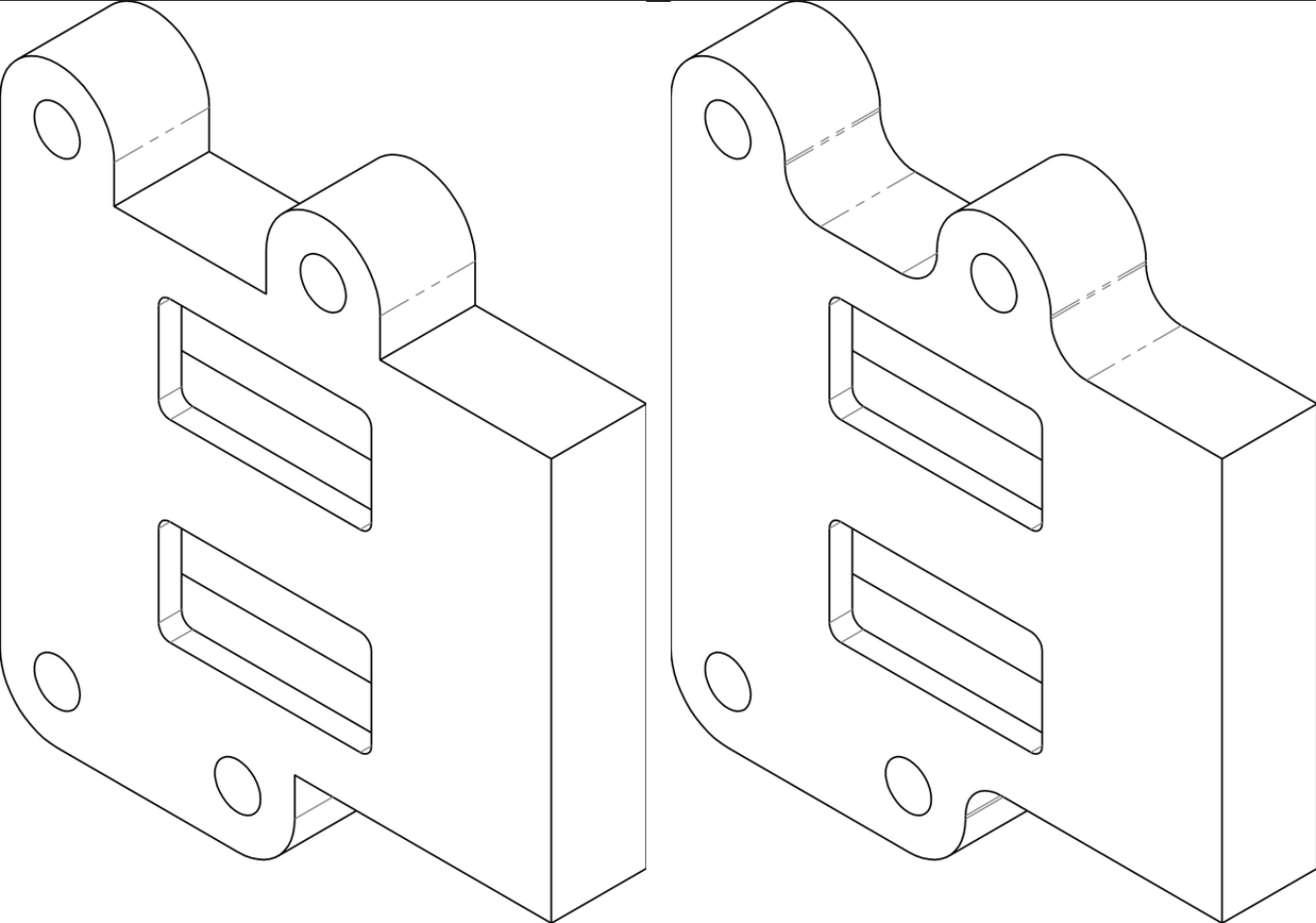
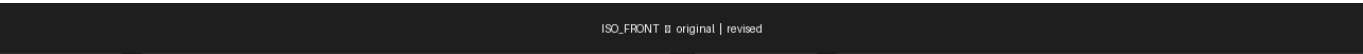


Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	annotation	16X R.030	8X R.030	0.92	The count of R.030 radii on the rectangular pockets changed from 16X to 8X, indicating either a reduction in the number of radiused corners or a re-classification of which radii are called out under this leader, affecting machining requirements. Impact: Halving the called-out R.030 radius count from 16 to 8 may reduce machining operations or indicate re-annotation; machinist must verify all pocket corners are accounted for to avoid under-machining.
MINOR	note	TAP FOR #4-40 HELICOIL	TAP FOR #4-40 HELICAL INSERTS	0.95	The insert type callout was changed from 'HELICOIL' (brand name) to 'HELICAL INSERTS' (generic term), which is a documentation/terminology change that does not affect the functional specification. Impact: Purely a documentation/terminology change from brand name to generic — no functional or manufacturing impact expected.

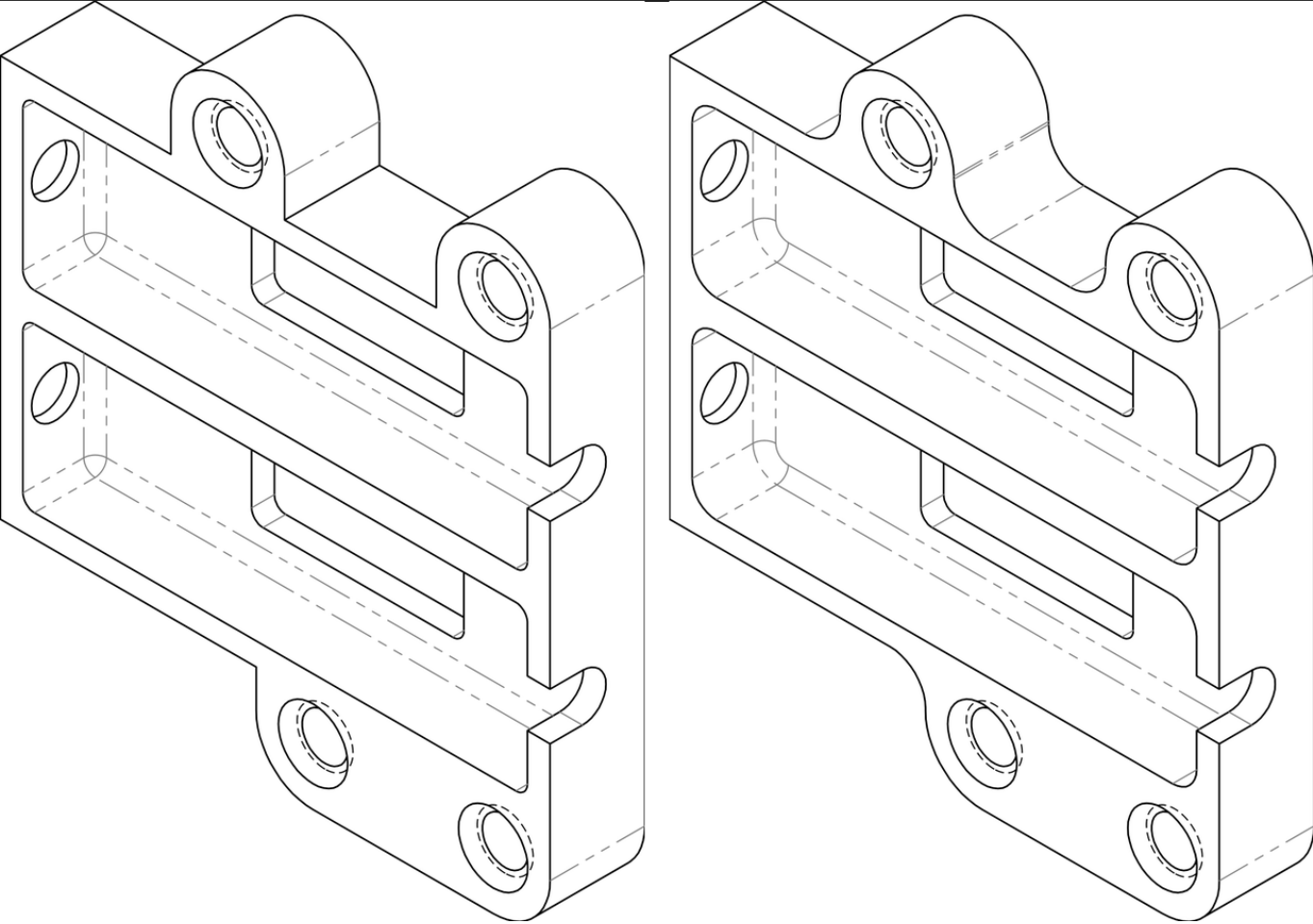
ANNOTATION [MATCHED]



Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	geometry	Flat/straight bottom edge profile on front isometric view	Rounded/filleted bottom edge profile on front isometric view	0.72	The lower corners/edges of the bracket in the front isometric view appear to have added fillet radii, changing the part geometry from sharp/straight to rounded profiles. Impact: Added fillets on the bracket's lower edges change the machining envelope and may affect fixturing, deburring steps, and mating-part clearance.
MINOR	annotation	REV NC, INITIAL RELEASE & SEE EO., 06/28/25	REV A, SEE EO, 10/16/25	0.95	Revision table updated from NC (initial release) to Rev A with new date and simplified description. Impact: Administrative revision history update only — no manufacturing or functional impact.
MINOR	note	P/N: XXXXX-1 REV: NC	P/N: XXXXX-1 REV: A	0.95	Note 3 identification marking updated from REV NC to REV A to match the new revision level. Impact: Identification marking note updated to match new revision level — documentation only, but part marking/labeling must be updated accordingly.

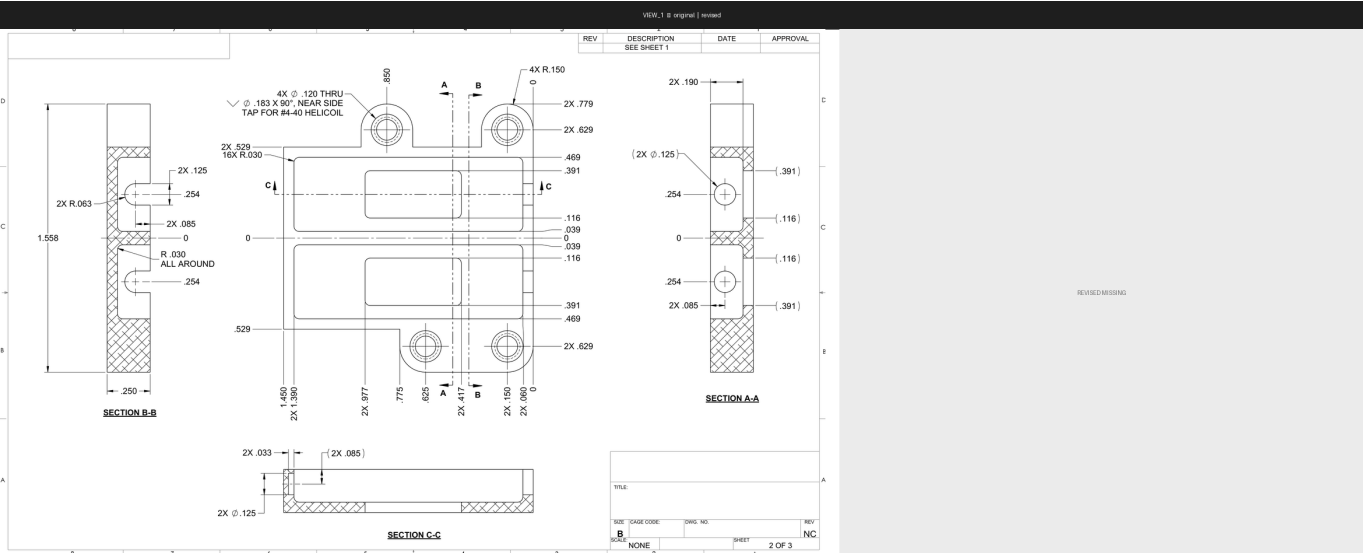


Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	geometry	Left mounting boss/tab has a straight/angular transition from the cylindrical roll to the flat face	Left mounting boss/tab has a larger, more pronounced curved (S-shaped) transition from the cylindrical roll to the flat face	0.72	The left upper mounting lug shows a changed fillet/blend geometry where the rolled cylindrical feature transitions to the main body, indicating a modified radius or swept profile that affects the part shape and potentially its strength. Impact: Modified blend geometry on the left mounting lug changes the swept profile, requiring updated CAM toolpaths and potentially affecting structural load transfer at that feature.



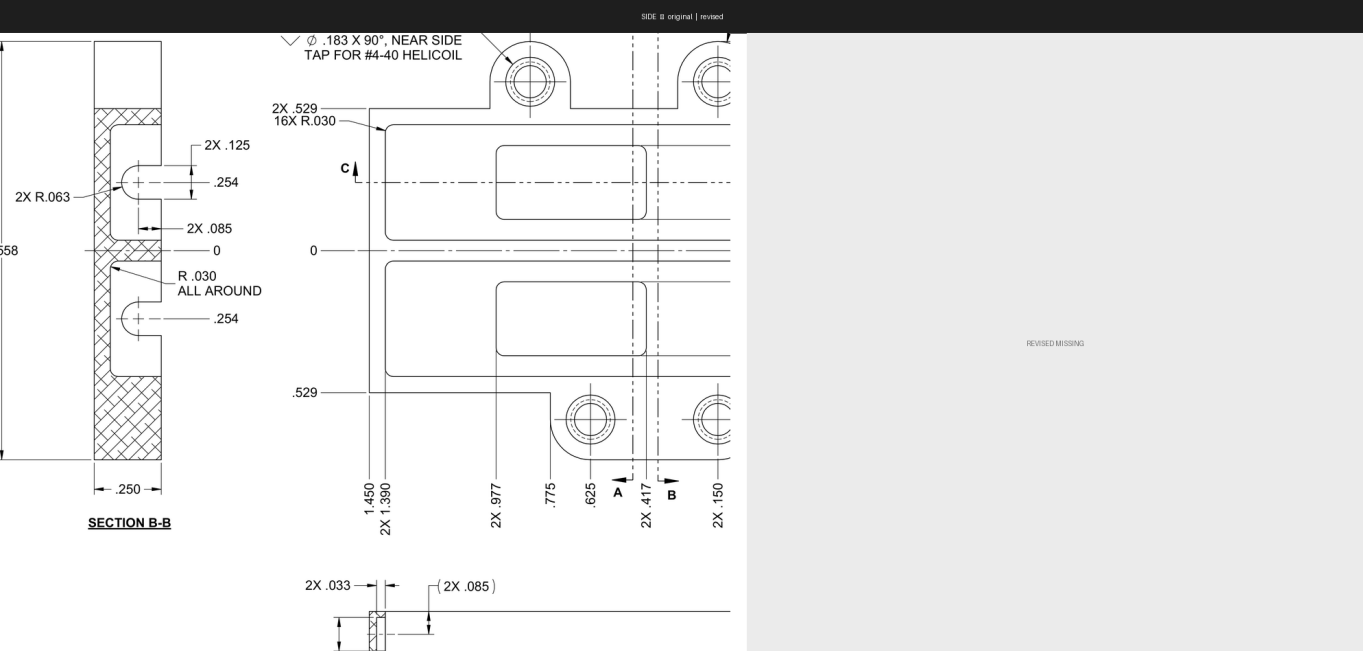
Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	geometry	Sharp interior corners at left-side vertical edges of the two rectangular slot openings (upper and lower slots on left face)	Filleted/radiused interior corners at left-side vertical edges of the two rectangular slot openings (upper and lower slots on left face)	0.75	The revised view shows curved/ filleted transitions at the interior corners of the left-side slot openings where the original had sharp corners, indicating a geometry change that affects stress concentration and potentially manufacturing method. Impact: Adding fillets to previously sharp slot interior corners reduces stress concentration and changes end-mill requirements; confirm fillet radius is explicitly dimensioned elsewhere on the drawing.

VIEW_1 [REMOVED]



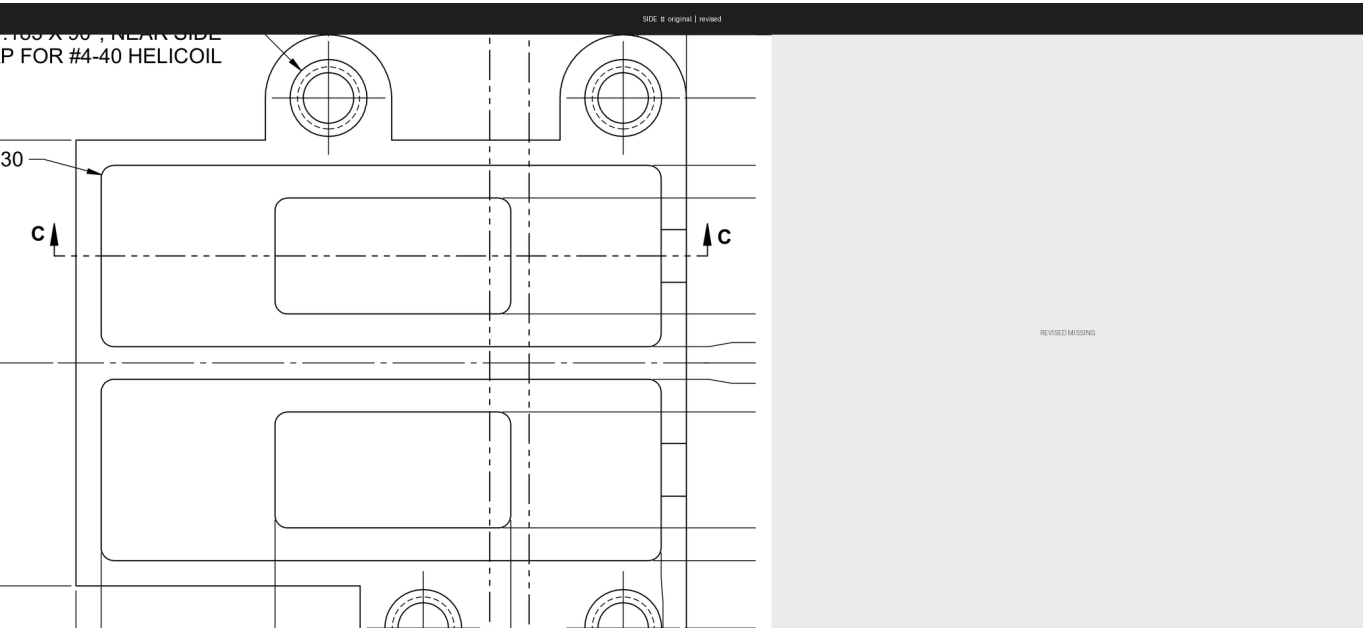
Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	view	VIEW_1	—	1.00	View 'VIEW_1' was present in the original but not the revised drawing. Impact: VIEW_1 removed from drawing set — verify that all dimensions and features formerly shown in this view are fully captured in the new view structure.

SIDE [REMOVED]



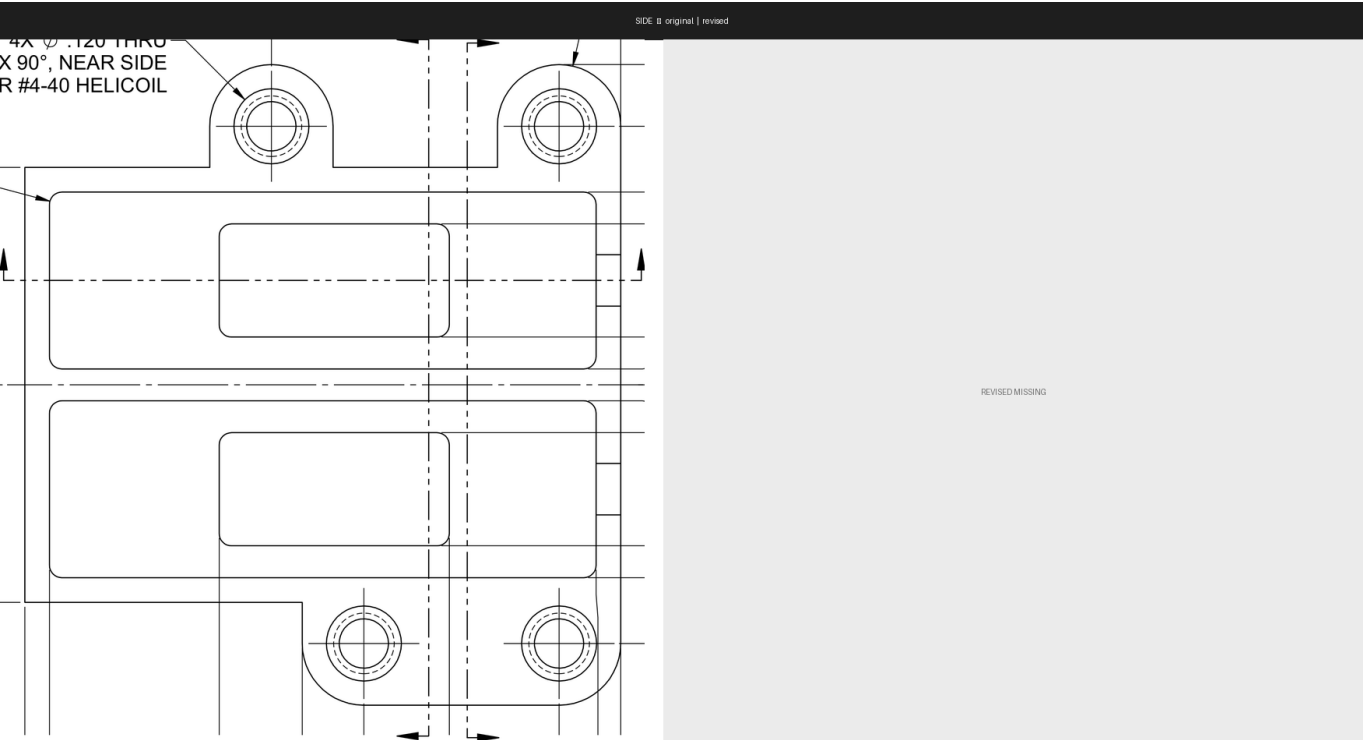
Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	view	SIDE	—	1.00	View 'SIDE' was present in the original but not the revised drawing. Impact: SIDE view removed — confirm all side-face dimensions and tolerances are re-expressed in the replacement section views to avoid underdefined geometry.

SIDE [REMOVED]



Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	view	SIDE	—	1.00	View 'SIDE' was present in the original but not the revised drawing. Impact: SIDE view removed — confirm all side-face dimensions and tolerances are re-expressed in the replacement section views to avoid underdefined geometry.

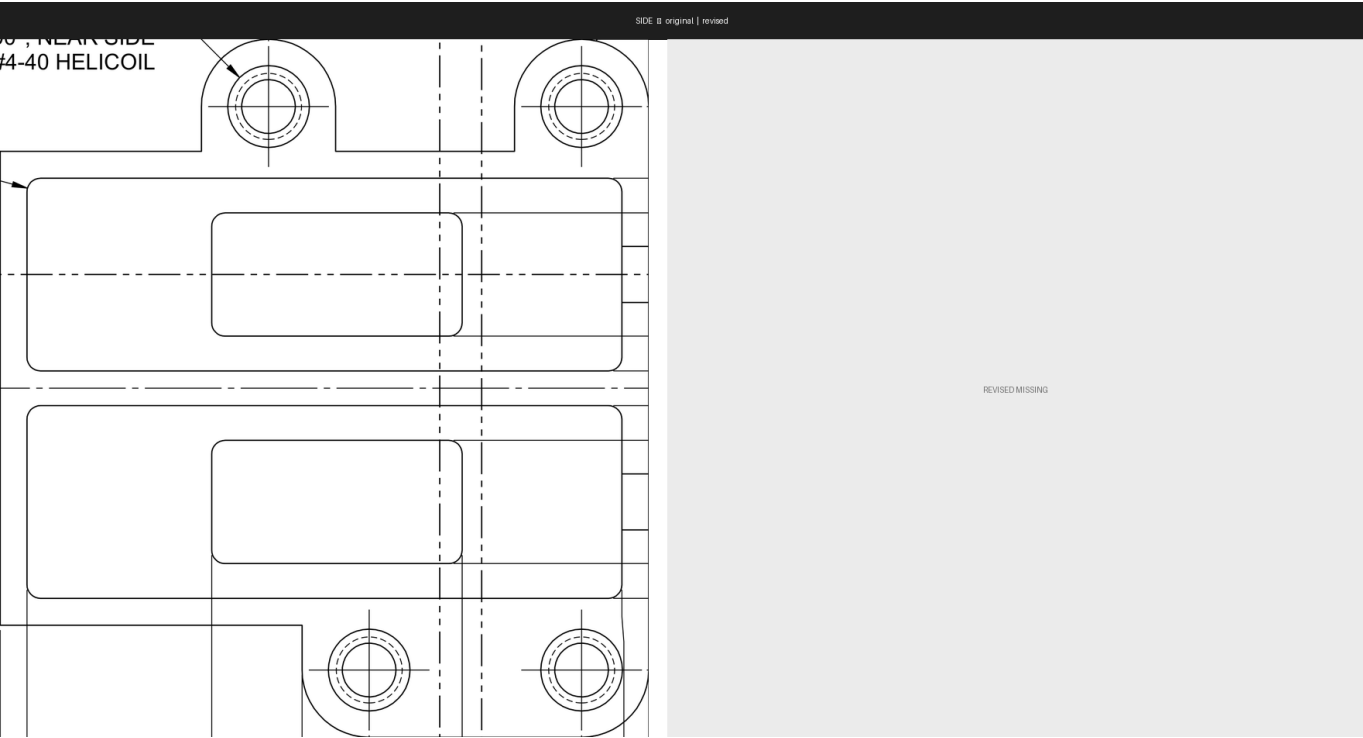
SIDE [REMOVED]



Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	view	SIDE	—	1.00	View 'SIDE' was present in the original but not the revised drawing.

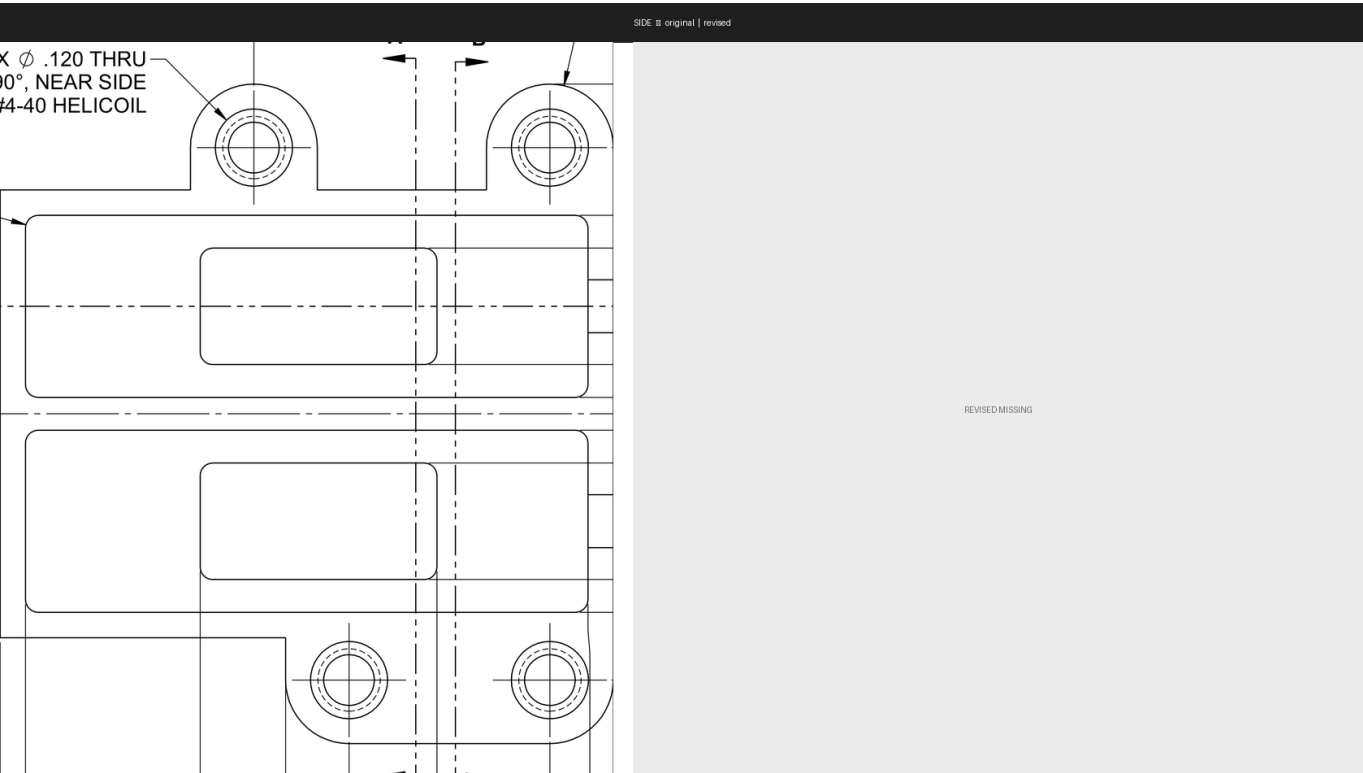
Severity	Field	Original	Revised	Conf.	Rationale
					Impact: SIDE view removed — confirm all side-face dimensions and tolerances are re-expressed in the replacement section views to avoid underdefined geometry.

SIDE [REMOVED]



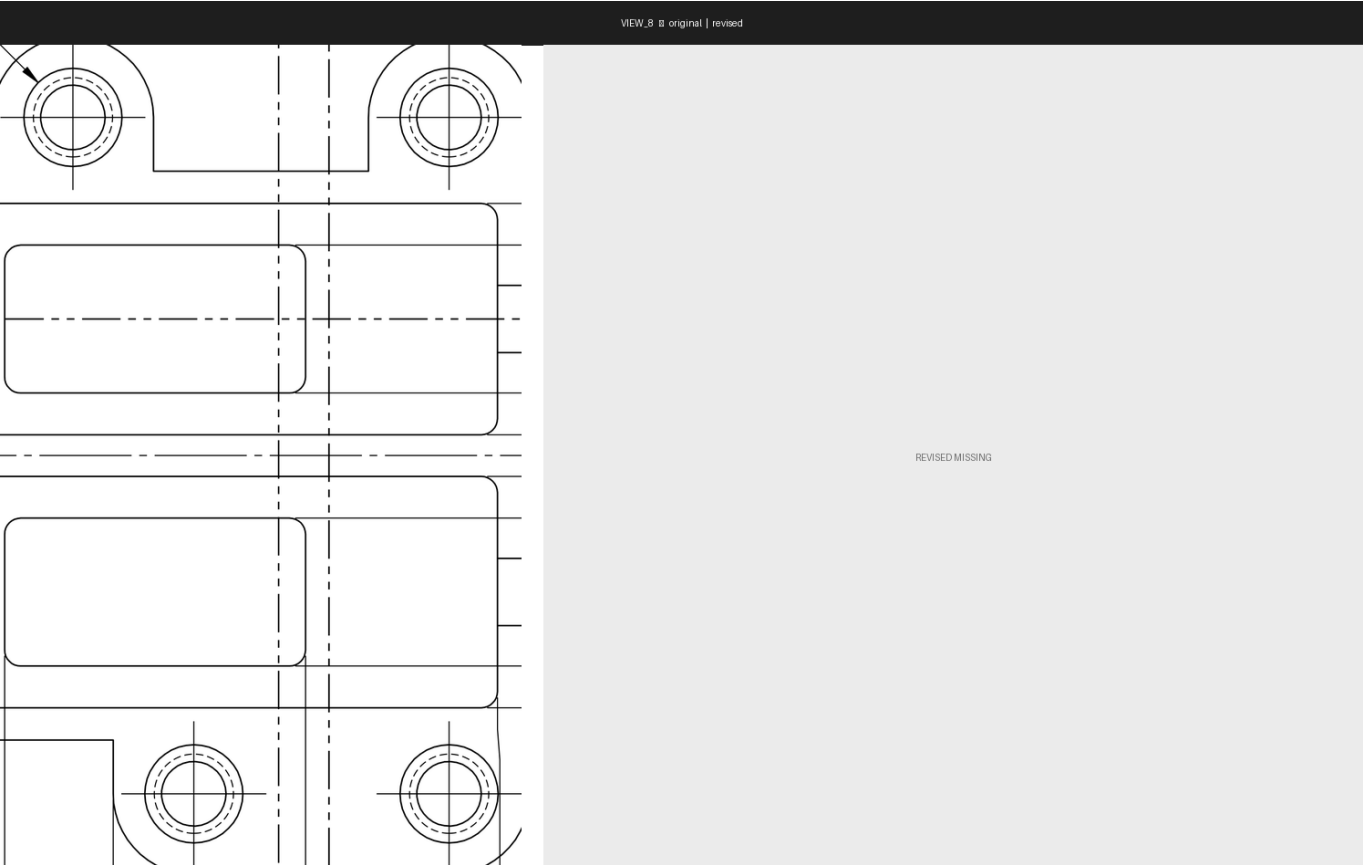
Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	view	SIDE	—	1.00	View 'SIDE' was present in the original but not the revised drawing. Impact: SIDE view removed — confirm all side-face dimensions and tolerances are re-expressed in the replacement section views to avoid underdefined geometry.

SIDE [REMOVED]



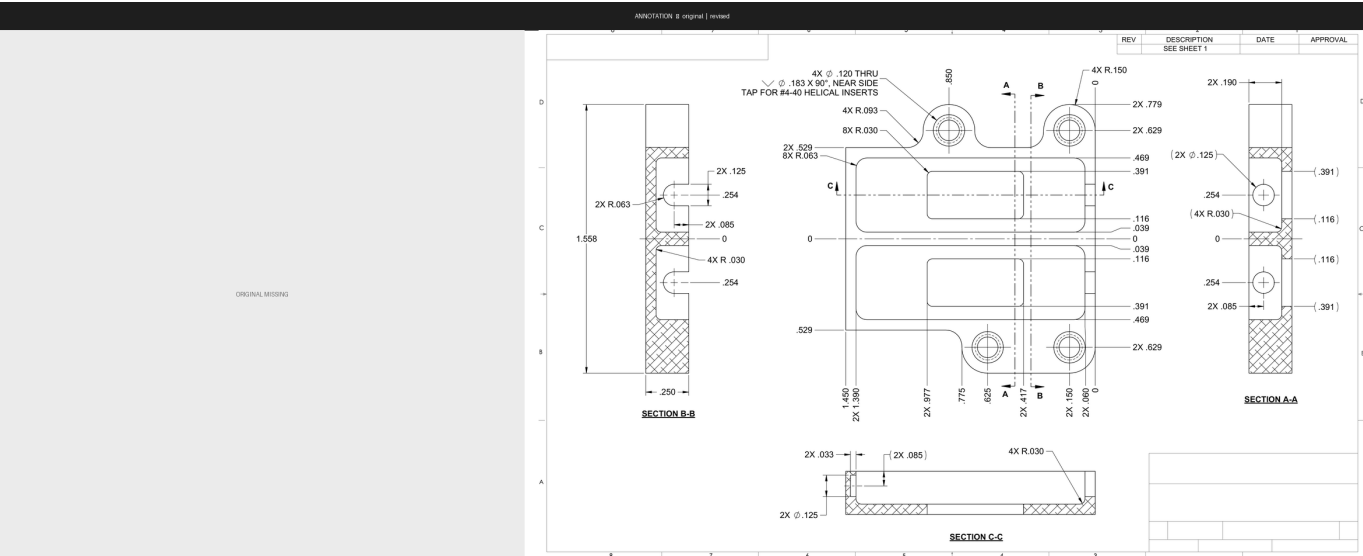
Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	view	SIDE	—	1.00	View 'SIDE' was present in the original but not the revised drawing. Impact: SIDE view removed — confirm all side-face dimensions and tolerances are re-expressed in the replacement section views to avoid underdefined geometry.

VIEW_8 [REMOVED]



Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	view	VIEW_8	—	1.00	View 'VIEW_8' was present in the original but not the revised drawing. Impact: VIEW_8 removed — reviewer should confirm no unique dimensions or notes from this view were lost in the drawing restructure.

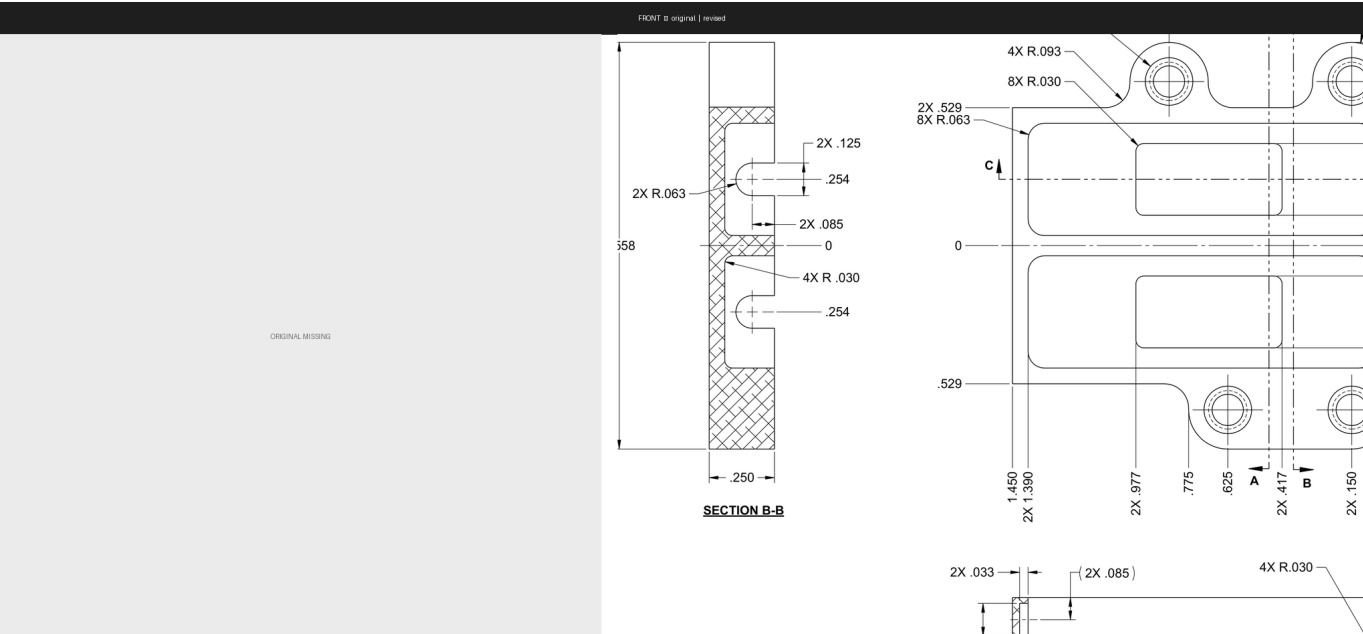
ANNOTATION [ADDED]



Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	view	—	ANNOTATION	1.00	View 'ANNOTATION' appears only in the revised drawing.

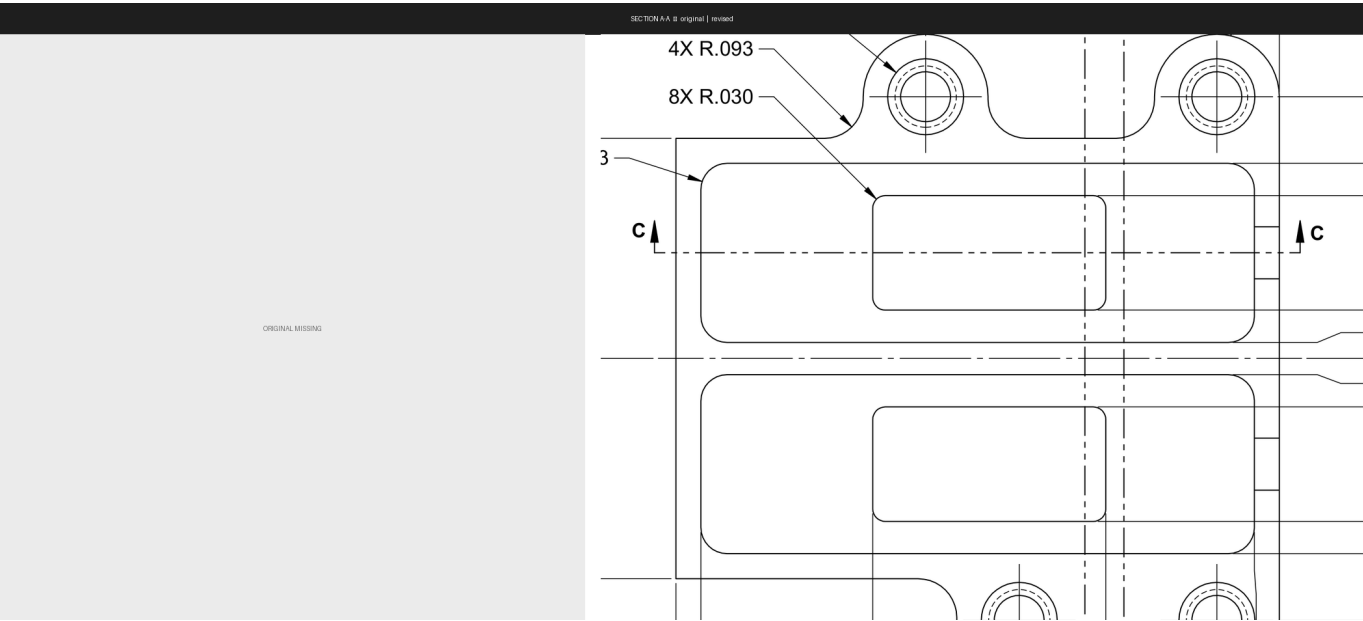
Severity	Field	Original	Revised	Conf.	Rationale
					Impact: New ANNOTATION view added to consolidate notes and revision data — documentation restructuring, no direct manufacturing impact but review for completeness.

FRONT [ADDED]



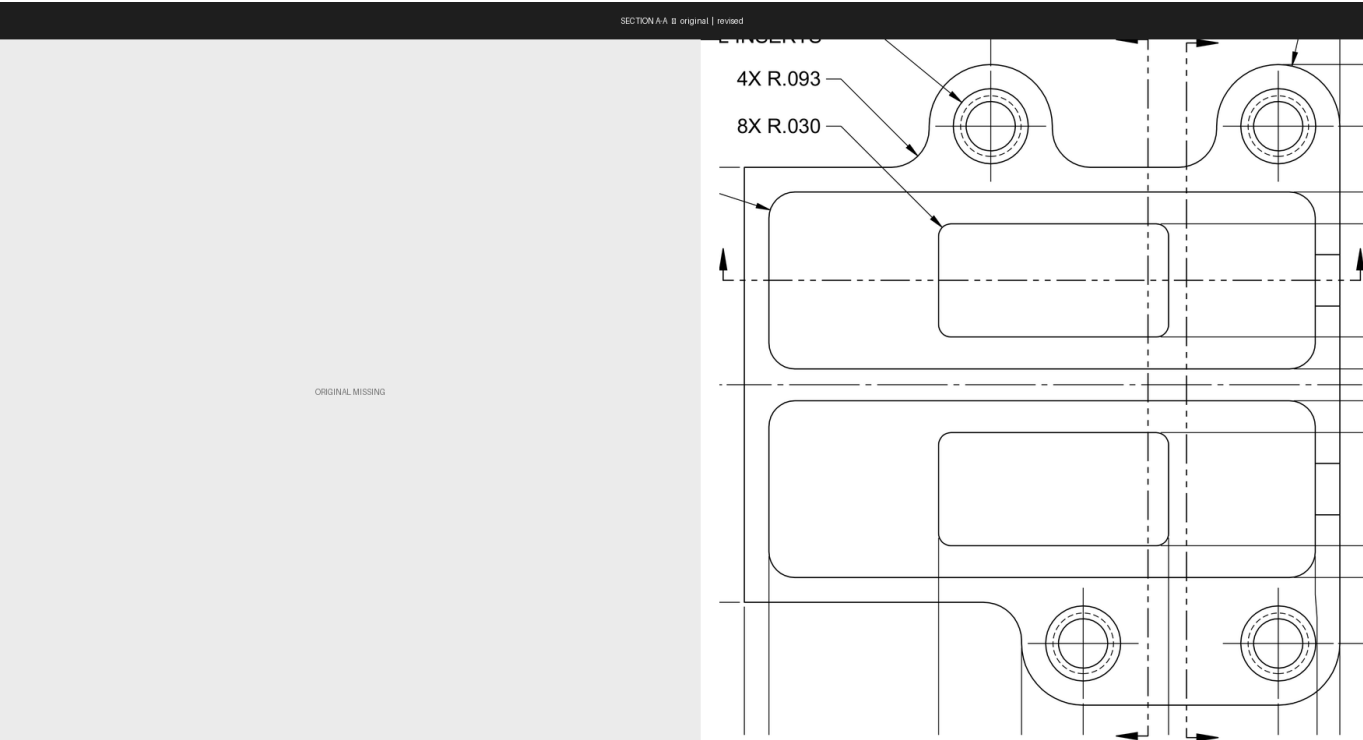
Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	view	—	FRONT	1.00	View 'FRONT' appears only in the revised drawing. Impact: New orthographic FRONT view added, providing explicit 2D geometry reference that replaces isometric-only coverage — improves manufacturing clarity.

SECTION A-A [ADDED]



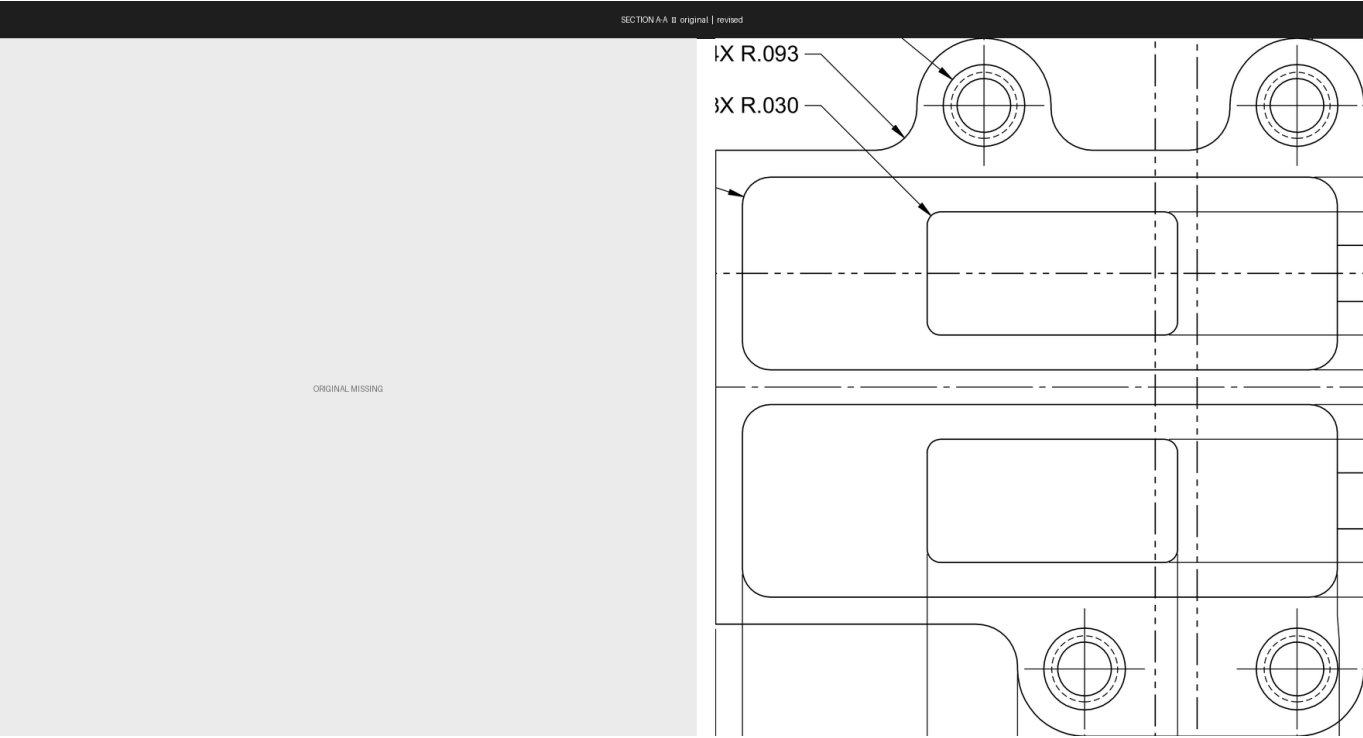
Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	view	—	SECTION A-A	1.00	View 'SECTION A-A' appears only in the revised drawing. Impact: New Section A-A added to the revised drawing, exposing internal features not previously sectioned — machinist and inspector should use this view for hole/pocket depth verification.

SECTION A-A [ADDED]



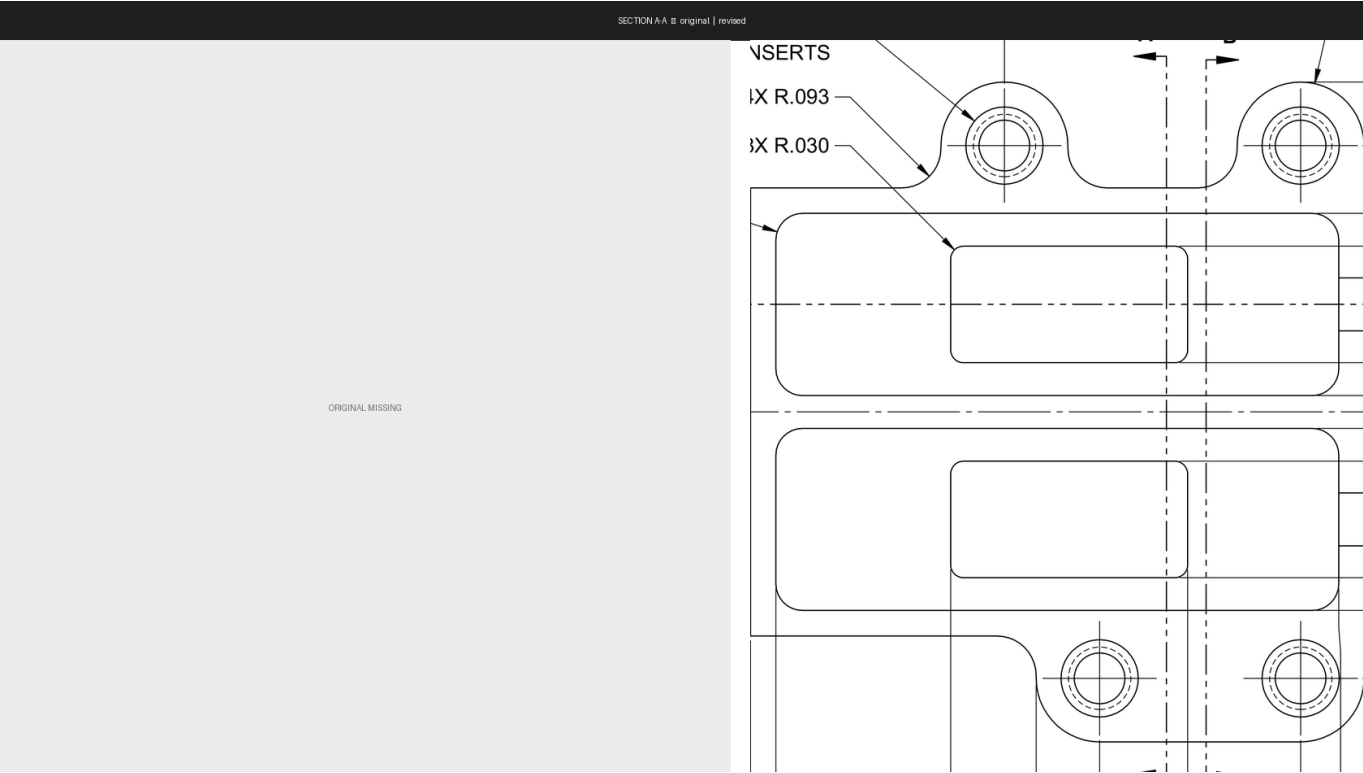
Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	view	—	SECTION A-A	1.00	View 'SECTION A-A' appears only in the revised drawing. Impact: New Section A-A added to the revised drawing, exposing internal features not previously sectioned — machinist and inspector should use this view for hole/pocket depth verification.

SECTION A-A [ADDED]



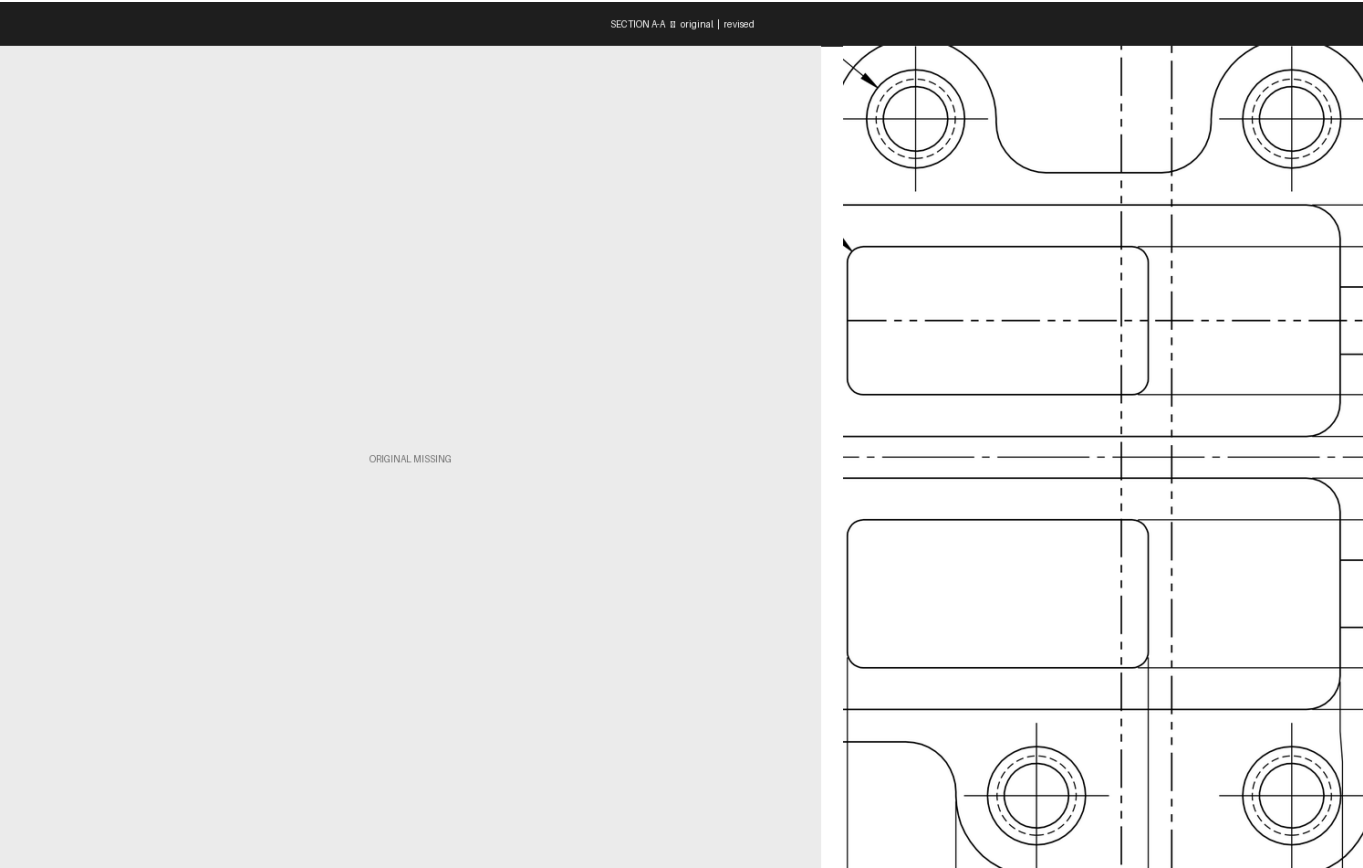
Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	view	—	SECTION A-A	1.00	View 'SECTION A-A' appears only in the revised drawing. Impact: New Section A-A added to the revised drawing, exposing internal features not previously sectioned — machinist and inspector should use this view for hole/pocket depth verification.

SECTION A-A [ADDED]



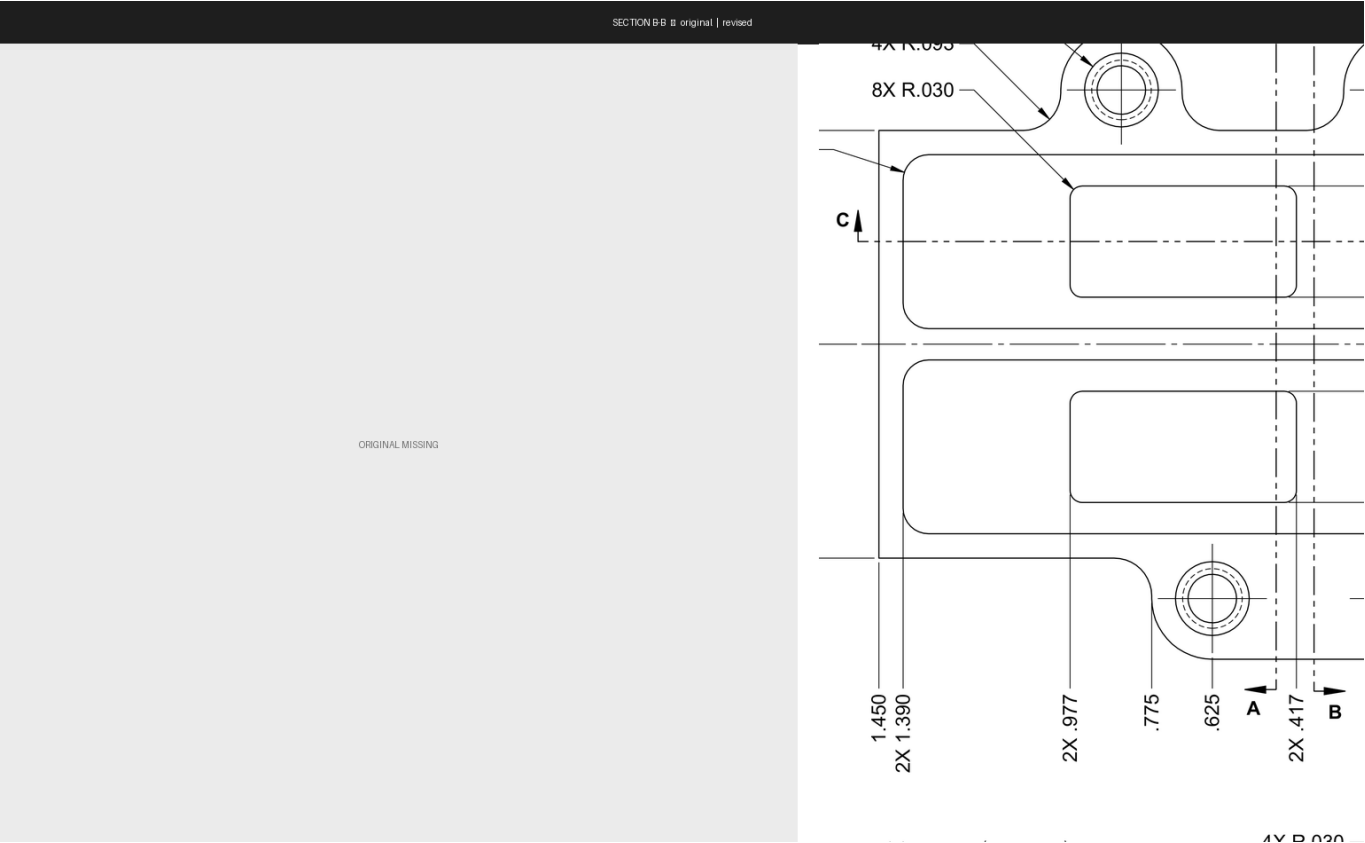
Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	view	—	SECTION A-A	1.00	View 'SECTION A-A' appears only in the revised drawing. Impact: New Section A-A added to the revised drawing, exposing internal features not previously sectioned — machinist and inspector should use this view for hole/pocket depth verification.

SECTION A-A [ADDED]



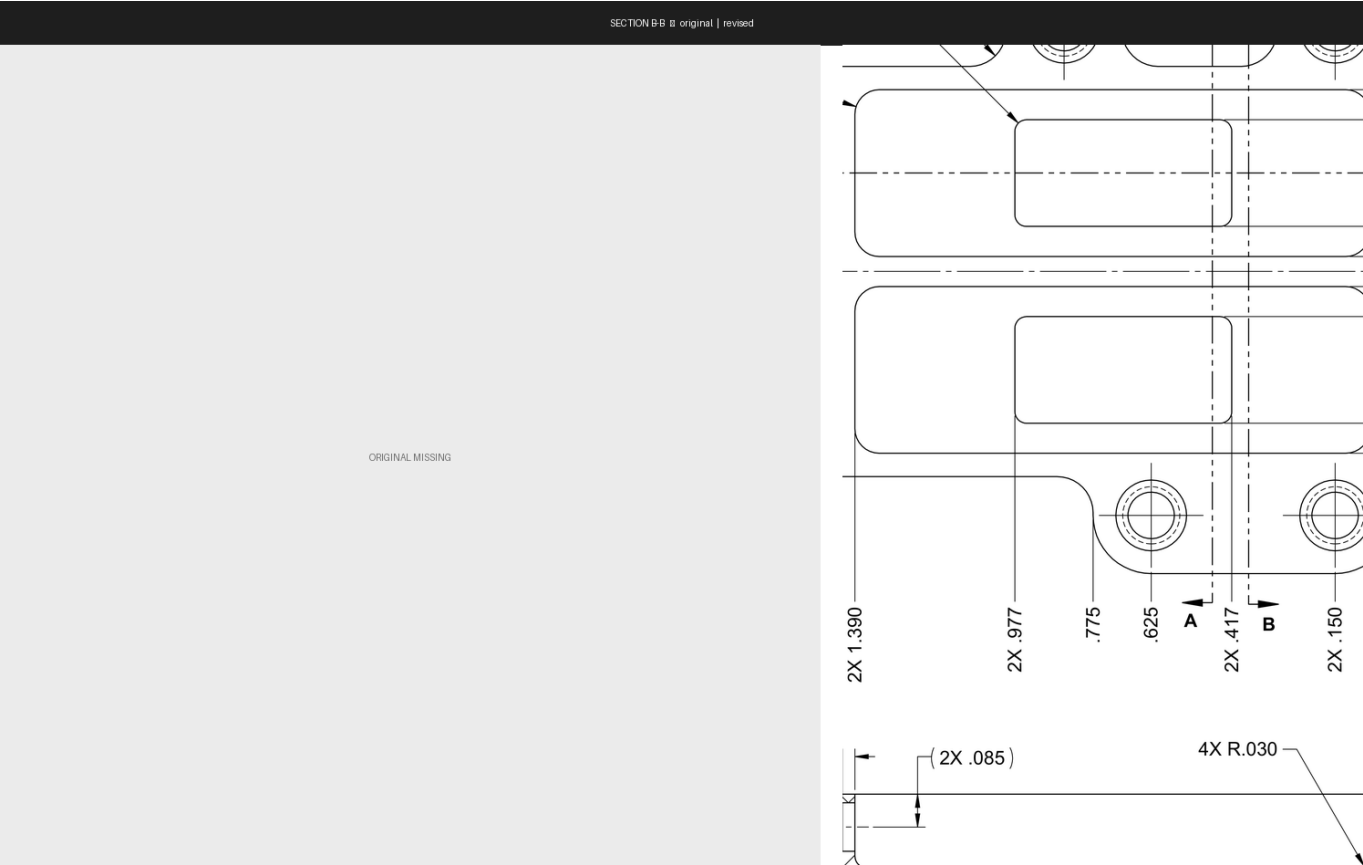
Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	view	—	SECTION A-A	1.00	View 'SECTION A-A' appears only in the revised drawing. Impact: New Section A-A added to the revised drawing, exposing internal features not previously sectioned — machinist and inspector should use this view for hole/pocket depth verification.

SECTION B-B [ADDED]



Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	view	—	SECTION B-B	1.00	View 'SECTION B-B' appears only in the revised drawing. Impact: New Section B-B added, providing additional cross-sectional detail; confirm all dimensions in this section are consistent with mating part interfaces.

SECTION B-B [ADDED]



Severity	Field	Original	Revised	Conf.	Rationale
SIGNIFICANT	view	—	SECTION B-B	1.00	View 'SECTION B-B' appears only in the revised drawing. Impact: New Section B-B added, providing additional cross-sectional detail; confirm all dimensions in this section are consistent with mating part interfaces.

Legend

- **CRITICAL** — Affects structural integrity, safety, interchangeability, or regulatory compliance.
 - **SIGNIFICANT** — Affects manufacturing scope, cost, or lead time. No safety risk but requires re-planning.
 - **MINOR** — Clarity, formatting, or documentation change. No manufacturing impact.
 - **UNCERTAIN** — Confidence below 0.65. Requires human review.
- ⚠ **AI-generated** — requires human verification before release to manufacturing.